# Inequality in the Context of a Mechanical Model of Society

*Nature, Nurture, and the Separation of Mechanism from Correction*

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## Abstract

Inequality can rise even when efforts to moderate it are making a substantial difference. This paper examines how accumulation and correction interact through a mechanical baseline in which heterogeneous endowments and accumulated outcomes reinforce one another. Starting from equal outcomes, the baseline produces a rising mean and increasing inequality. A budget-balanced transfer example shows why equalizing current stocks need not prevent renewed divergence: inequality subsequently rises, but remains substantially below the uncorrected outcome. We distinguish that corrective effect from the further improvement attainable under the available interventions. The latter comparison requires an efficient frontier defined for a specified mechanism, correction technology, timing, and horizon. Together, the baseline and frontier organize questions about what correction changes, what continues to generate inequality, and what further combinations of mean and inequality are achievable. The purpose is to make correction intelligible without losing sight of the continuing process it modifies.

**Keywords:** inequality; cumulative advantage; mechanical baseline; correction; efficient frontier; nature and nurture

## 1. Introduction

Suppose inequality rises after an effort to reduce it. Has the effort failed? Not necessarily. Inequality might have risen further without it. Nor does stable inequality tell us that little is happening: a continuing tendency toward divergence and a continuing correction can leave an apparently unchanged result. The observed distribution is the outcome to be explained, not an explanation of how much either process contributes.

We begin with the intuition that advantage can compound. Resources already accumulated can help their holders acquire more, while differences in capacity or access affect how strongly that feedback operates. When gains are retained, the pursuit of larger individual outcomes can produce both a growing aggregate and a widening distribution. The recurring question is why inequality can seem to be on a one-way train even when deliberate efforts are made to change its course.

Nature and nurture provide a familiar way to approach this interaction. A tree grows through its own processes, while cultivation changes the form that growth takes. Here, “nature” names the operation of a specified cumulative mechanism; “nurture” names interventions that modify its outcomes or operation. The distinction is relative to a reference setting, whose institutions can shape both accumulation and the corrections available.

We first construct a baseline society in which the cumulative mechanism operates without the specified corrections. We then introduce correction and ask which combinations of mean outcome and inequality

become attainable. Among those combinations, the efficient frontier identifies outcomes with no further improvement available in one dimension without a sacrifice in the other. Comparing a corrected outcome with the baseline asks what correction has accomplished; comparing it with the frontier asks what else is achievable.

The persistence and renewal of substantial inequality give this question its historical motivation. The World Inequality Report 2026, for example, documents substantial global wealth concentration in 2025 and considerable differences across regions [1]. Such patterns prompt questions about the processes sustaining concentration and the interventions modifying them. Establishing an uncorrected comparison requires a separate account of those processes and interventions.

Section 2 makes the baseline explicit. Section 3 follows one balanced transfer through its immediate and subsequent effects and derives a frontier for the worked example. Section 4 uses that example to distinguish correction effects from remaining possibilities, while Section 5 asks what different corrections change in the process. Sections 6 and 7 develop questions for application and their connections with earlier and contemporary work. Throughout, growth and distribution remain parts of the same inquiry.

## 2. The Baseline Society

### 2.1 The expectation and its implementation

Our baseline society has two defining features: its aggregate outcome grows, and the distribution of that outcome becomes more unequal. These are two aspects of the same process. Members retain their gains, and members who are better positioned to generate gains increase their outcomes faster.

We select the model to implement a historically motivated expectation and explore its consequences. Its results characterize that construction rather than supply independent evidence for the expectation. Cumulative advantage has an established intellectual history: Merton examined the disproportionate allocation of recognition to already established scientists [2], and DiPrete and Eirich review processes through which a favorable position becomes a resource for further gains [3]. Piketty's account of capital and inheritance provides another route into the persistence and renewal of concentration [4]. These accounts motivate attention to feedback while leaving its mathematical representation to be specified.

The historical record also contains compression. Goldin and Margo document a narrowing of the US wage structure during the 1940s [5]. To explain a departure from a diverging baseline, an application must identify what changed: the accumulation process, the interventions, the population, or the conditions under which outcomes were measured. We isolate the cumulative process here so that the consequences of introducing a correction can be followed directly.

### 2.2 Members, outcomes, and the growth rule

Consider a fixed population of $n \geq 2$ members. Each holds a positive quantity $y_i(t)$ of one accumulated outcome at integer time $t$. This outcome is a retained resource stock, measured in common units, that can support further accumulation. Wealth is a possible interpretation; annual income, consumption, and well-being are not interchangeable with this stock. The mean is the amount per member, not a complete measure of social welfare.

Members start with equal outcomes, $y_i(0) = c > 0$, but have different fixed, positive endowments $e_i$. We use *endowment* to summarize a member's capacity to convert available resources into additional outcome in the reference environment. It can represent acquired capacity or access as well as individual characteristics; it is not a measure of moral worth or deservingness.

We hold the population and institutional environment fixed. Members remain in the population indefinitely, and the baseline excludes birth, death, inheritance, consumption, depreciation, and randomness. These assumptions isolate retained gains and their subsequent accumulation from changes in population, institutions, or other channels.

We represent that accumulation through the following growth rule:

$$y_i(t+1) = y_i(t)[1 + \rho e_i y_i(t)] = y_i(t) + \rho e_i y_i(t)^2. \tag{1}$$

Here $\rho > 0$ controls feedback strength, with units chosen so that $\rho e_i y_i(t)$ is dimensionless. The proportional growth rate increases with both endowment and current outcome. This is a strong version of cumulative advantage: a larger stock increases not only the next increment but also its proportional rate. The fixed rule carries each period's result into the next, making the process mechanical and self-moving. It represents accumulation without separately solving members' decisions or assigning society a single conscious goal.

The rule uses one outcome variable and one fixed endowment per member, together with one common feedback parameter. We examine its operation over stated finite horizons. Each finite step produces finite outcomes, although the trajectories grow without bound as the number of steps increases; the model therefore supplies a dynamic benchmark rather than a stationary long-run distribution.

### 2.3 Growth and divergence together

At the first step, the difference between two members' outcomes is $\rho c^2(e_j - e_i)$. Equal starting outcomes therefore do not prevent inequality from emerging. Thereafter, the higher-endowment member also holds a larger stock, and the two advantages reinforce one another.

We track the mean alongside relative inequality, measured by the population Gini coefficient:

$$\mu(t) = \frac{1}{n}\sum_{i=1}^{n} y_i\,(t), \qquad G(t) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n}\left|y_i(t) - y_j(t)\right|}{2n^2\mu(t)}. \tag{2}$$

A common proportional increase in all outcomes leaves this measure unchanged. For a population of $n$ members its upper bound is $(n-1)/n$; for two members that bound is $1/2$. All Gini values below use this convention.

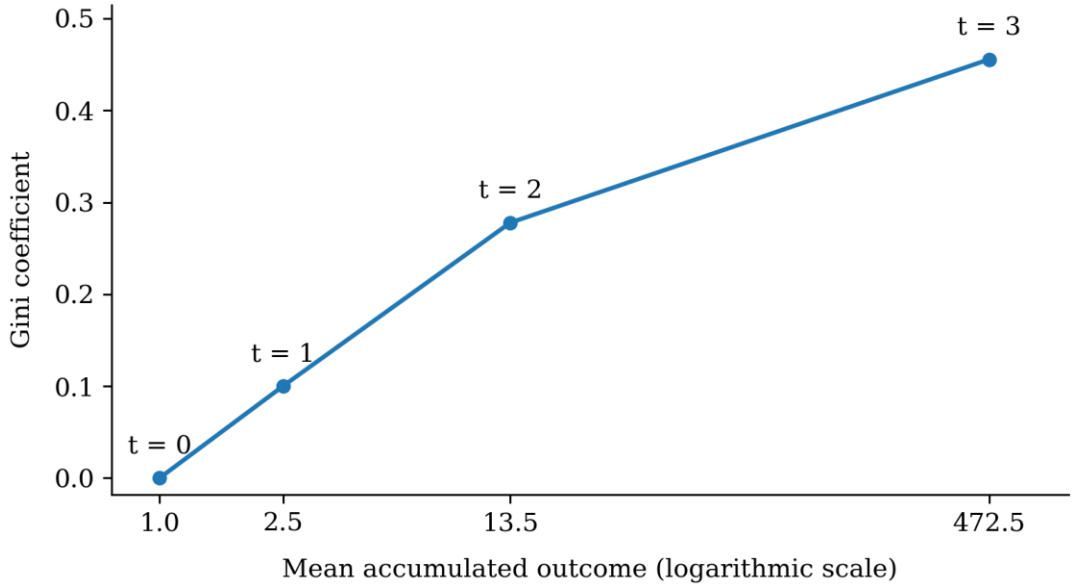
**Baseline result.** Under Equation (1), equal positive starting outcomes and heterogeneous positive endowments produce a strictly rising mean and a strictly increasing Gini coefficient. More strongly, every lower cumulative population share receives a smaller share of the total at the next step. Appendix A.1 supplies the argument.

The reason is straightforward. Ordering members by endowment also orders their outcomes. At each step, the growth multipliers are larger toward the top of that ordering. Everyone gains in absolute terms, but the lower part of the distribution receives a declining share of the growing total. This is the one-way train

within the baseline: starting from equal outcomes, endowment heterogeneity initiates divergence, and the stock-dependent growth rate then reinforces it. With identical endowments and equal starting outcomes, equality would persist.

For a running example, take two members with $c = 1$, $e_1 = 1$, $e_2 = 2$, and $\rho = 1$. Their outcomes are $(1,1)$ initially, $(2,3)$ after one step, and $(6,21)$ after two. The mean rises from 1 to 2.5 to 13.5, while the Gini rises from 0 to 0.1 to approximately 0.2778. These are deliberately illustrative units and periods, not calibrated economic quantities.

**Figure 1. Mean growth and increasing inequality on the uncorrected path**




Note: Equation (1), $c = 1$, $(e_1, e_2) = (1,2)$, $\rho = 1$. The points follow one uncorrected trajectory across integer time steps. The horizontal axis is logarithmic because the baseline accelerates rapidly. Values are model calculations.

## 2.4 The positional reading

We can also read the baseline through the outcomes of individual members. A member's outcome reflects the endowment carried into the process and the gains retained from its earlier operation. Holding the environment fixed makes this relationship visible without changes in institutions or luck obscuring it.

The distributional reading asks how these individual trajectories fit together. In the running example, the first member's stock rises from 2 to 6 while its share of the total falls from two-fifths to two-ninths. The member remains lower-ranked; it is the share, rather than the rank, that declines. Individual gains and increasing relative inequality can therefore occur together. The positional and distributional readings describe the same accumulation process from different perspectives.

# 3. The Frontier

## 3.1 What becomes achievable under correction?

The baseline answers what happens without the specified intervention. Introducing correction raises a different question: what combinations of mean and inequality can be attained? The answer depends on what the correction can do, when it acts, and when outcomes are evaluated.

We use *correction technology* to mean the available intervention rules, their resource requirements, and their timing. For a fixed starting distribution, endowment profile, feedback rule, population, and horizon, these interventions generate an *attainable set* of mean–inequality pairs. One attainable pair dominates another if it has a mean at least as high and inequality no greater, with at least one strict improvement. The nondominated pairs form the *efficient frontier*. Efficiency here concerns these two coordinates; a broader welfare assessment may require additional dimensions.

This definition turns a desired combination into a question about feasibility. Once the mechanism and available corrections are specified, we can ask whether that combination is attainable and whether it can be improved. Different starting conditions or intervention arrangements can yield different frontiers. We therefore derive the frontier together with the correction technology, rather than infer it from the general presence of cumulative advantage.

## 3.2 A balanced transfer before the next growth step

Consider a correction that moves every outcome a fraction $\tau$ toward the current mean, where $0 \leq \tau \leq 1$. Write the outcomes immediately after the transfer as $z_i(t)$:

$$z_i(t) = (1-\tau)y_i(t) + \tau\mu(t), \qquad \sum_{i=1}^{n}[z_i(t) - y_i(t)] = 0. \tag{3}$$

Members above the mean contribute; members below it receive; members at the mean are unchanged. Total contributions equal total receipts. At $\tau = 1$, everyone holds the current mean, and no transfer reverses the ranking of outcomes. Immediately after the transfer, the mean is unchanged and the Gini is $(1-\tau)G(t)$.

The next growth step applies the same mechanism to the corrected stocks:

$$y_i(t+1) = z_i(t) + \rho e_i z_i(t)^2. \tag{4}$$

The transfer itself neither creates nor destroys resources. Any subsequent change in the mean relative to the uncorrected path comes from the different stocks entering the growth rule. This timing separates the direct compression from its dynamic consequences.

Within this technology, $\tau$ controls the degree of compression. We vary that intensity while keeping the growth rule, endowments, and measurement date fixed, so the resulting differences can be traced to the transfer.

## 3.3 A derived frontier in the two-member example

The sequence is clearest at full compression. The two members hold 2 and 3 units. Transferring half a unit leaves each with 2.5, so inequality is zero immediately after the transfer. Their different endowments then

produce next-period outcomes of 8.75 and 15. Equalizing the stocks has changed the resources entering accumulation, but the higher-endowment member still generates the larger gain.

To trace all intensities, return to the outcomes $(2,3)$ reached after the first baseline step. The correction transfers $\tau/2$ from the second member to the first, giving stocks $(2+\tau/2,\ 3-\tau/2)$. Applying Equation (4) produces:

$$y_1' = 6 + \frac{5}{2}\tau + \frac{1}{4}\tau^2, \qquad y_2' = 21 - \frac{13}{2}\tau + \frac{1}{2}\tau^2. \tag{5}$$

The resulting next-period mean and Gini are:

$$\mu(\tau) = 13.5 - 2\tau + \frac{3}{8}\tau^2, \qquad G(\tau) = \frac{15 - 9\tau + \tau^2/4}{54 - 8\tau + 3\tau^2/2}. \tag{6}$$

Over $0 \le \tau \le 1$, both the next-period mean and the next-period Gini decrease strictly. As the transfer increases, the first member's next outcome rises and the second member's falls; their relative gap contracts. Meanwhile, less stock enters the higher-return branch of this particular growth rule, reducing the next-period total. Here the member holding more also has the higher conversion capacity, as the equal-start baseline produces. The mean effect of compression depends on that alignment. Appendix A.2 establishes the result for the corresponding aligned two-member case.

**Table 1. One balanced transfer, followed by one growth step**

| Transfer intensity $\tau$ | Transfer amount | Next outcomes $(y_1', y_2')$ | Next mean | Next Gini |
|---|---|---|---|---|
| 0 | 0.000 | (6.0000, 21.0000) | 13.5000 | 0.2778 |
| 0.5 | 0.250 | (7.3125, 17.8750) | 12.5938 | 0.2097 |
| 1 | 0.500 | (8.7500, 15.0000) | 11.8750 | 0.1316 |

*Note:* Initial stocks $(2,3)$, endowments $(1,2)$, feedback $\rho = 1$. The transfer precedes growth; no second transfer occurs before measurement. Values are rounded model calculations.

Every point on this curve is efficient within the stated family: obtaining a lower Gini requires a lower next-period mean, and obtaining a higher mean requires a higher Gini. The baseline is the high-mean endpoint for this family. Maximal correction is the low-Gini endpoint, but it does not deliver equality at the measurement date, because unequal endowments generate fresh divergence after the transfer.

The final row answers the opening question. Inequality rises from its initial value of $0.1$ to $0.1316$, even though the correction has reduced it substantially relative to the uncorrected value of $0.2778$. The mean also continues to grow, from $2.5$ to $11.875$, while ending below the uncorrected mean. Correction has changed the course of accumulation without eliminating either growth or the conditions for renewed divergence.

### 3.4 Why timing belongs in the frontier

Now change only the transfer's timing. Allow the uncorrected stocks $(2,3)$ to grow to $(6,21)$ and then apply Equation (3), toward their mean of $13.5$. The transfer leaves that mean unchanged at every intensity. Full compression gives each member $13.5$ units and produces equality, so the efficient set for

this terminal-transfer technology consists of the pair $(13.5,0)$ when only terminal mean and Gini are assessed. If both transfer timings are admitted as options under these terminal objectives, the fully equalized terminal outcome dominates every point on the pre-growth curve.

The difference arises because the two schedules put different stocks through the growth step before measurement. The pre-growth transfer changes those stocks; the terminal transfer reallocates the outcome already produced. This comparison evaluates final stocks and excludes behavioral responses to anticipated transfers, administrative costs, intermediate welfare, and subsequent growth. A policy comparison including those considerations would need to specify them in the attainable set. The trade-off in Section 3.3 thus belongs to the mechanism, transfer timing, and horizon used there.

Optimal-taxation analysis likewise connects attainable outcomes to objectives, information, incentives, and available instruments [6]. Those ingredients determine what a proposed correction can achieve and what sacrifices it entails. With the relevant frontier specified, we can turn from the choice of an intervention to the interpretation of its outcome.

# 4. The Classification

## 4.1 Location, objectives, and comparability

To classify an outcome, we record the combination of mean and inequality attained, its difference from a matched baseline, and the further improvement attainable under the available corrections. These entries describe different aspects of the result. In the running example, the same corrected outcome can be compared with the uncorrected path to assess its effect and with other feasible outcomes to assess what remains possible.

A stated objective is recorded separately. On the frontier in Table 1, a stronger preference for lower inequality can favor a larger transfer, whereas a higher-mean target can favor a smaller one. Observing the chosen intensity does not establish that preference: the outcome may also reflect power, imposed constraints, or incomplete implementation. Evidence about the objective is therefore distinct from the coordinates of the outcome.

Comparisons require a common basis. We align the outcome concept, population, units, inequality measure, starting conditions, and horizon, and identify relevant differences in the mechanism and correction technology. Real societies can be compared relative to their own frontiers or under an explicitly standardized benchmark. Empirical use requires evidence for the selected baseline and feasible interventions; the classification specifies how those comparisons fit together.

## 4.2 Two comparisons that answer different questions

The first comparison holds the starting conditions and horizon fixed and asks what correction changed relative to the uncorrected path. In Table 1, full pre-growth correction reduces the next-period Gini by approximately $0.1462$ and changes the mean by $-1.625$. Both belong in the account of the correction's effect. The second comparison starts from the corrected outcome and asks what further improvement is available.

For an attainable outcome with mean $\mu_o$ and Gini $G_o$, define its remaining mean-improvement gap as:

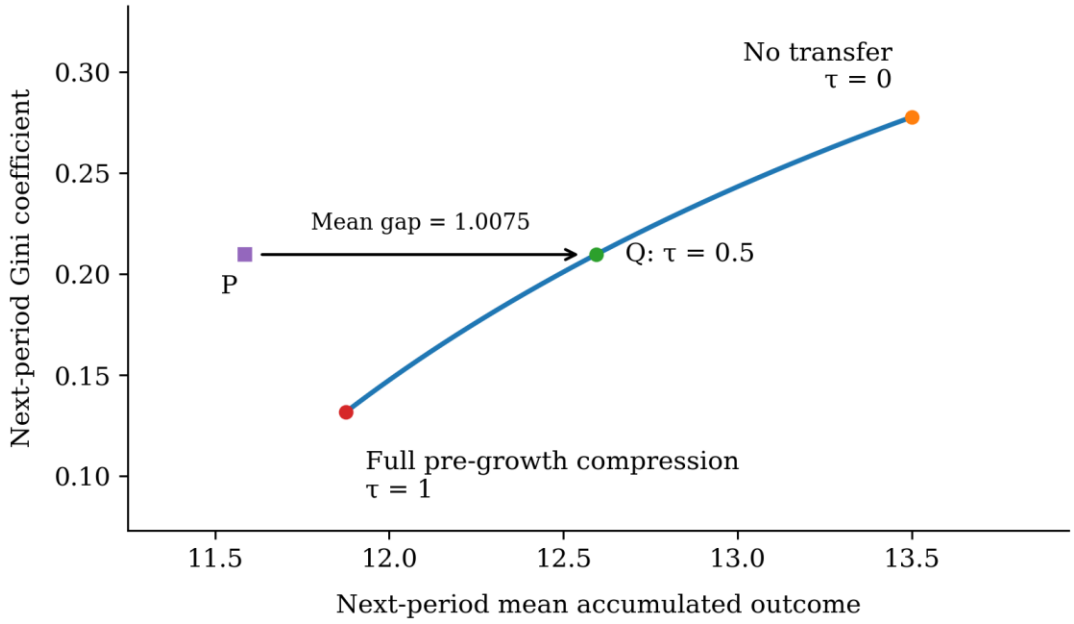
$$D_\mu = \sup\{\mu: (\mu, G) \text{ is attainable and } G \leq G_o\} - \mu_o. \quad (7)$$

This gap measures how much higher the mean could be without increasing inequality. It is expressed in outcome units. An analogous inequality gap, expressed in Gini units, holds the mean at least as high. Both comparisons matter: a zero mean gap alone need not establish efficiency, since a lower Gini at the same mean could still be attainable.

Consider the middle row of Table 1, labeled Q in Figure 2. Suppose an avoidable proportional loss after growth removes 8% of each member's outcome, producing P, while Q remains feasible without that loss. For this comparison we include such proportional losses among the attainable outcomes. They add dominated points without changing the loss-free frontier. At P, the mean is 11.5863 and the Gini remains about 0.2097.

Relative to the uncorrected baseline, P has a lower Gini by about 0.0681 and a mean of 11.5863 rather than 13.5. Relative to Q, a further 1.0075 in mean outcome is attainable without increasing its Gini. The first comparison records what the implemented correction accomplished; the second records what avoidable loss prevented it from accomplishing. Figure 2 places the two outcomes against the same frontier.

**Figure 2. The frontier for pre-growth transfers and a remaining-improvement comparison**




Note: The curve is given by Equations (5)–(6). Q uses $\tau = 0.5$; P has an additional proportional loss of 8%. The arrow measures attainable mean improvement at the same Gini when loss-free implementation is feasible. Values are model calculations.

The improvement from P to Q is feasible because the loss is avoidable in this example. In an application, binding resource or institutional constraints must be included in the attainable set. A shortfall relative to a less constrained comparator may instead represent a difference in capability. Likewise, costs involving

health, security, or other objectives need to be included in a broader assessment before a two-coordinate improvement is treated as a welfare improvement.

### 4.3 Movement along, toward, and of the frontier

The running example gives three movements distinct meanings. Choosing another transfer intensity in Table 1 moves *along* the fixed frontier, changing the combination of next-period mean and inequality. Removing the avoidable loss at P moves *toward* that frontier, reaching Q without increasing inequality. Making a previously unavailable transfer schedule feasible can change the attainable set and hence the frontier *itself*, as the timing comparison in Section 3.4 illustrates.

The comparison date also matters. A pre-growth transfer changes the stocks entering accumulation while leaving the growth parameters fixed. At the original starting date and horizon, choosing that transfer selects an outcome from the defined attainable set. At a later date, the resulting stocks become the starting conditions for a new comparison. An intervention can therefore affect later possibilities as well as the outcome initially evaluated.

Changing parameters and changing the available choice set are separate distinctions. If a parameter-changing intervention is already included among the admissible options, choosing it selects an outcome within that set. A change in what is available, or a new comparison with different fixed conditions, requires a newly specified frontier. Additional capabilities may expand the set; changed parameters can improve some combinations and worsen others. The direction follows from the resulting possibilities, not from the intervention's label.

## 5. Design Implications

### 5.1 What does the correction change?

We examine corrections by asking which part of accumulation they change. In the running example, a transfer changes the stocks entering the next growth step. A change in effective endowments changes how members convert resources into further gains. A change in feedback changes how strongly existing stocks reinforce subsequent accumulation. Improved delivery can remove avoidable losses, as in the move from P to Q.

These questions connect with Atkinson's treatment of interventions involving technology, market arrangements, capital, taxation, and social protection [7]. One intervention can affect several channels. An education intervention, for example, requires an account of whose capacity changes, how its costs are financed, and how the changed capacities affect later outcomes. Table 2 separates the questions needed to follow those effects.

**Table 2. Questions to ask of a correction**

| Point of intervention | Representation in the framework | Question the analysis must answer |
|---|---|---|
| Accumulated outcomes | A transfer or other change in $y_i$ | What changes immediately, and what stocks enter the next growth step? |
| Effective endowments | A specified change in $e_i$ | Whose capacity changes, by how much, at what cost, and for how long? |
| Feedback or access | A change in $\rho$ or the growth rule | Does the change alter the strength or distribution of subsequent gains? |
| Correction capacity | Different timing, coverage, constraints, or implementation losses | Which additional combinations become attainable after costs are included? |

Note: An intervention may affect several channels. Analysis of shocks, depreciation, or intergenerational transmission requires the corresponding extension to the baseline.

The table directs attention from the name of an intervention to its place in the accumulation process. We call changes in current stocks *state corrections*, and changes in parameters, rules, or available capabilities *structural corrections*. A transfer can have continuing effects because its corrected stocks feed into later accumulation. A structural change can alter subsequent gains for as long as the changed conditions persist. Assessing durability therefore requires following both what changes immediately and what remains changed in later periods.

## 5.2 Why a one-time equalization can be temporary

Recall what happened after the two members were brought to equal stocks of 2.5. Their endowments remained different, so the next growth step produced unequal gains. To generalize that result, suppose a transfer brings every member to the same stock $m > 0$. After one uncorrected growth step, outcomes become $m + \rho e_i m^2$. Let $\overline{e}$ be mean endowment and $G_e$ its Gini. The resulting inequality is:

$$G_{\text{next}} = \frac{\rho m \overline{e}}{1 + \rho m \overline{e}} \, G_e. \tag{8}$$

Equation (8) measures the inequality regenerated after equalization. It is positive when endowments differ, and stronger feedback produces greater next-step inequality from the same equalized stock and endowment profile. The transfer removes current outcome inequality while leaving heterogeneous conversion capacities in place. Changing those capacities or the feedback rule could therefore change the correction needed later; the resulting mean and distribution would both need to be assessed.

### 5.3 Matching correction to changing pressure

Regeneration raises a further question: how much correction is needed to reach a specified inequality target? At a given date, apply the mean-compression rule after growth and assess the target immediately after transfer. Write the pre-transfer Gini as $G^-$ and the target as $G^* \geq 0$. When $G^- > 0$, the minimum intensity required is:

$$\tau_{\min} = \max\left\{0,\ 1 - \frac{G^*}{G^-}\right\}. \tag{9}$$

For a fixed positive target below pre-transfer inequality, a higher pre-transfer Gini requires a greater intensity of compression. This follows from the immediate effect of the rule: the transfer multiplies the Gini by one minus its intensity. Under repeated intervention, the pre-transfer distribution must be evaluated along the corrected path, including the effects of earlier transfers. Equation (9) gives the required intensity at each comparison date. The rate must also be distinguished from the quantity transferred. A constant rate can move a growing quantity of resources as stocks increase.

A capacity limit makes the comparison sharper. If the largest feasible intensity is $\overline{\tau} < 1$, the target is unattainable through this rule from the given state whenever $(1 - \overline{\tau})G^- > G^*$. Reaching it would require a change in available correction or in the conditions generating inequality; otherwise the target would have to be reconsidered. The relevant question is whether corrective capacity can meet the target under the pressure actually being generated.

Current work on transformative AI gives these questions a concrete setting. Growiec, Prettner, and Szkróbka examine universal basic income, universal basic capital, and several tax instruments in a theoretical framework for automation and distribution [8]. In our terms, comparing those instruments requires distinguishing changes in ownership and resource stocks from changes in subsequent gains and ongoing redistribution. Their costs, timing, and persistence determine which comparisons would be informative.

## 6. Predictions and Questions for Application

### 6.1 What the model establishes

The baseline and transfer example establish conditional results: the specified accumulation process raises the mean and relative inequality; pre-growth compression in the worked family trades a lower next-period mean for lower inequality; and unequal endowments regenerate divergence after stocks have been equalized.

To turn these results into empirical hypotheses, an application must specify what outcomes and endowments represent, assess the growth mechanism, identify the intervention, and state when its effects are measured. That makes it possible to investigate whether the process operates as proposed and whether correction changes its course. The remaining-improvement gap has a different role: it organizes a comparison once the attainable set has been established.

### 6.2 Technology: amplification through a specified channel

The first technology question is how a change affects the connection between existing resources and subsequent gains. Access to productive tools could strengthen that connection when access depends on accumulated resources. Changes in task requirements could instead alter which capacities matter and who can use them. An extension can represent these channels through $\rho(A)$ or $e_i(A)$, where $A$ denotes a specified technology condition. The functions must describe the channel before its distributional effect can be assessed.

Labor-market outcomes provide one part of the evidence. Using job postings and labor-market data, Jaumotte and colleagues distinguish new skills generally from AI-related new skills. In their 2026 IMF study's US local-labor-market analysis, demand for new skills is associated with wage and employment gains concentrated among higher- and lower-skilled workers. AI-specific demand shows no statistically significant overall employment gain and is associated with employment losses in highly exposed occupations where AI has limited scope to complement workers [9].

Access and task requirements point to other channels. In a 2024 study of Latin America and the Caribbean, Gmyrek, Winkler, and Garganta examine how digital-access gaps can restrict potential productivity benefits from generative AI [10]. Althoff and Reichardt's 2026 task-based model instead finds an equalizing mechanism through simplification of task requirements, enabling workers with fewer relevant skills to enter previously inaccessible work [11].

These studies concern labor-market associations, potential occupational exposure, and modeled general-equilibrium effects. Their different objects help distinguish channels an application would need to investigate. To apply the present stock-accumulation model, we would also need to establish how changes in work or access affect retained resources and subsequent gains. Where independently assessed feedback becomes stronger, Equation (8) predicts greater next-step divergence from the same equalized stocks and fixed endowments. Equation (9) then gives the correction required for a specified pre-transfer Gini and target under the compression rule.

### 6.3 Accumulation time, resets, and comparison

Elapsed time matters because retained gains become resources for further accumulation. Along the same uncorrected trajectory, later distributions are more unequal than earlier ones. Equalizing current stocks resets outcome inequality but leaves the heterogeneous endowments that generated different gains. A subsequent growth step therefore begins a new round of divergence.

An empirical comparison should follow the duration and interruptions of the relevant accumulation process. Calendar age alone does not identify that history. Comparing populations at the same mean also requires attention to starting stocks, endowments, feedback, and interventions, because the mean rises with time along the fixed baseline. When a disruption changes several of these conditions, the new starting point and accumulation rule must be identified before the later outcome is compared with an earlier path.

### 6.4 From observed outcomes to a usable comparison

An application begins by fixing the outcome, population, and horizon, then specifying the accumulation process to be examined. Observations can help assess that process and constrain its parameters. Evidence

about intervention timing, coverage, costs, and implementation can then support an uncorrected comparison and delimit the feasible alternatives. Both components must remain open to revision when the evidence conflicts with them.

With those comparisons in place, the analyst can report what correction changed and which improvements remain attainable, while recording any stated objective separately. A before-and-after change supplies part of the outcome history; identifying a corrective effect additionally requires the uncorrected comparison. Assessing remaining possibilities additionally requires the attainable set. These are the steps that turn "Why is inequality still high?" into distinguishable questions about cumulative pressure, correction, and what could be done next.

## 7. Historical and Contemporary Context

The connection between growth and distribution places this inquiry alongside several traditions in political economy. Smith examined productive improvement, the distribution of its proceeds, and the accumulation of stock within the same inquiry [12]. Ricardo placed the division of the product among rent, profit, and wages at the center of political economy [13]. Marx analyzed concentration and centralization within his account of capitalist accumulation [14]. These different accounts make the distribution of a growing product part of the question about accumulation itself.

The cumulative-advantage literature focuses more closely on the feedback through which a favorable position becomes a resource for later gains [2,3]. That connection explains why a snapshot of the distribution is incomplete for the present purpose: the resources it records also enter the process producing the next distribution. Our mechanical construction follows that connection explicitly and lets an intervention change what enters the next step.

A second connection concerns the evaluation of the resulting combinations. Pigou treated the size and distribution of the national dividend as related dimensions of economic welfare [15]. Optimal-taxation analysis links objectives to incentives, information, and instruments [6]. These connections distinguish two tasks: establishing which combinations are attainable and deciding which combination to pursue. Mean and inequality help describe the alternatives; the choice between efficient combinations requires an objective and, where relevant, additional dimensions of welfare.

The population boundary determines whose outcomes enter that assessment. Milanovic distinguishes within-country from between-country inequality in examining global distribution [16]. Piketty's treatment of capital and inheritance gives intergenerational transmission an explicit role [4], while Atkinson examines a range of possible interventions [7]. These perspectives guide different applications: specify whose accumulation is followed, how gains are retained or transmitted, and which interventions alter the process. The two comparisons can then be maintained within the chosen setting rather than imposed on unlike populations or horizons.

## 8. Conclusion

Inequality's persistence does not reveal how little or how much correction has accomplished. A process can keep generating divergence while interventions substantially change its course. Conversely, a

correction can produce a meaningful improvement relative to an uncorrected path and still leave attainable improvements unrealized.

Our mechanical baseline shows how growth and increasing inequality can arise together when different endowments interact with retained outcomes. Equalizing current stocks changes the starting conditions for the next growth step, but leaves unequal capacities to generate further gains. The worked example therefore produces higher inequality than before correction, yet substantially less than on the uncorrected path. The correction matters because it changes the course of a process that continues.

A meaningful corrective effect is also compatible with further attainable improvement. The loss example makes this distinction visible: the intervention reduces inequality relative to the baseline, while better implementation can raise the mean without increasing inequality. Establishing that second possibility requires knowing what is feasible under the relevant timing, costs, and constraints.

Nature and nurture provide the intuitive starting point; following their interaction gives the comparison its content. Assessing correction requires asking what the process would have produced without it, what the intervention changed, and what further combinations remain attainable. Those questions distinguish a continuing problem from an ineffective response—and a meaningful improvement from the best available one.

# Appendix A. Mathematical Details

## A.1 Baseline growth and the distributional result

Order endowments as $e_1 \leq \cdots \leq e_n$. Equal positive starting outcomes and Equation (1) preserve the same ordering of outcomes, with strict separation wherever endowments differ. Every increment $\rho e_i y_i^2$ is positive, so the mean increases strictly. Each finite iterate is obtained by finite additions and multiplications; there is no finite-integer-time singularity. Moreover, $y_i(t) \geq c(1 + \rho e_i c)^t$, so every member's outcome is unbounded as $t \to \infty$.

Let $f_i = 1 + \rho e_i y_i$ be the multiplier at a given step. The $f_i$ are ordered with the $y_i$ and are not all equal. For a lower group consisting of members $1, \ldots, k$, its next cumulative outcome share is its current share multiplied by the ratio of its outcome-weighted mean multiplier to the population's outcome-weighted mean multiplier. For every $1 \leq k < n$, the former is strictly smaller. Hence each interior cumulative share declines. The piecewise-linear Lorenz curve lies strictly lower between its endpoints, so its area declines and the Gini increases. This establishes the stronger distributional result as well as the claim for Equation (2).

For comparisons of feedback strength, only the products $\rho e_i$ enter the growth rule. Comparisons across endowment profiles therefore require a common endowment scale, for example normalization to a population mean of one.

## A.2 A two-member frontier under pre-growth compression

Let the current outcomes satisfy $0 < a < b$ and the endowments satisfy $0 < e_1 \leq e_2$. Define $m = (a + b)/2$ and $d = (b - a)/2 > 0$. The corrected stocks are $z_1 = m - (1 - \tau)d$ and $z_2 = m + (1 - \tau)d$. Write $F_i(z) = z + \rho e_i z^2$ and $u_i = F_i(z_i)$. Both functions are strictly increasing in positive stocks.

As $\tau$ increases, $u_1$ increases and $u_2$ decreases. Since $u_2 \geq u_1 > 0$, the ratio $u_2/u_1$ decreases strictly, and so does $G = (u_2 - u_1)/[2(u_1 + u_2)]$. For the mean,

$$\frac{d\mu}{d\tau} = \rho d\,(e_1 z_1 - e_2 z_2) \leq 0. \tag{A1}$$

The derivative is negative throughout $0 \leq \tau < 1$; it is also negative at $\tau = 1$ if $e_2 > e_1$. The mean therefore decreases strictly over the interval. No two distinct transfer intensities dominate one another. Thus the entire attainable curve is the efficient frontier for this fixed pre-growth transfer family. Allowing additional uniform realization losses adds dominated points without changing that frontier.

For the general framework, if $\mathcal{C}$ is the specified set of admissible intervention schedules and $y^{\pi}(T)$ is the resulting distribution under schedule $\pi$, the attainable set is

$$\mathcal{A}_T = \left\{\left(\mu\left(y^{\pi}(T)\right), G\left(y^{\pi}(T)\right)\right) : \pi \in \mathcal{C}\right\}. \tag{A2}$$

Its efficient frontier is the subset satisfying the non-domination definition in Section 3.1. A nonempty compact set of schedules and a continuous finite-horizon outcome map ensure a compact attainable set and existence of efficient points. This is satisfied by the one-step example with $\tau \in [0,1]$. A general multi-period frontier is defined by the framework but is not solved by the one-step derivation.

### A.3 Direct compression and regeneration

Equation (3) preserves the mean and multiplies every pairwise absolute difference by $1 - \tau$. Substitution in Equation (2) gives $G(z) = (1 - \tau)G(y)$. This identity establishes the immediate effect of the transfer and Equation (9). If the pre-transfer Gini is zero, no correction is required to meet a nonnegative target.

From common stocks $m$, the next-step mean is $m + \rho m^2 \overline{e}$, while every pairwise absolute difference is $\rho m^2 |e_i - e_j|$. Substituting these quantities into Equation (2) gives Equation (8). At fixed $m$ and endowments, its derivative with respect to $\rho$ is $m\overline{e}G_e/(1 + \rho m\overline{e})^2$, which is positive when endowments differ.

## AI-Use Disclosure

AI tools were used in the development and revision of this working paper for exploratory discussion, literature discovery and checking, drafting assistance, mathematical analysis, numerical illustration, and editorial revision. The author selected the research question and directed the argument. Responsibility for reviewing and approving the final content remains with the author.

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